

PANYNJ Traffic & Bus Terminal Analytics Platform

Bridge, Tunnel & Bus Terminal Intelligence | Port Authority of New York & New Jersey

Project Overview

A full-scale, industry-sponsored analytics platform answering **6 real business questions** for the Port Authority of New York and New Jersey (PANYNJ). Built as a capstone project for the Data Analytics for Business course at the **University of New Haven (Spring 2026)**, analyzing:

- **5,383,378** bridge and tunnel lane-hour records (2013–2025)
- **1,601** bus terminal records across 3 facilities (2020–2025)

Using **8 production-ready models** across regression, classification, clustering, and time-series forecasting, delivering an estimated **\$23–40M in annual savings and revenue recovery opportunities**.

Key Results at a Glance

Metric	Value
Total traffic records analyzed	5.38M lane-hour observations
Total toll violations identified	81.1M (2013–2025)
EZ-Pass adoption rate	85.7% vs. 94% ISTHA benchmark
Holt-Winters forecast accuracy (MAPE)	2.89% — exceeds TfL 7% benchmark
H2O AutoML best model RMSE	703.81 (GBM)
2026 traffic forecast	127M vehicles
2027 traffic forecast	129M vehicles
2028 traffic forecast	132M vehicles
Congestion pricing traffic decline (2025)	58–62% across ALL facilities
H2O Classification best model error	0.00074 (GBM_5)
Models built and validated	8
Business questions answered	6
Estimated annual savings opportunity	\$23–40M

The Six Business Questions Answered

#	Business Question	Method	Key Finding
Q1	Top 5 factors affecting	Random Forest +	Type, Carrier, Year, Month, Quarter drive 98% of

#	Business Question	Method	Key Finding
	terminal usage	SHAP	volume. Speed variables = near-zero SHAP
Q2	Toll violation patterns & recovery	Trend + facility analysis	81.1M violations; GWB Lower tops at ~19M; peak 2020 (COVID); \$15–25M recovery opportunity
Q3	Busiest times & seasonality	Heatmap + hourly analysis	AM peak 06:00–09:00, PM peak 15:00–18:00; Friday highest day; January busiest month
Q4	Congestion pricing impact 2025	YoY regression analysis	All facilities down 58–62%; trip suppression confirmed — NOT diversion
Q5	2028 traffic volume forecast	H2O AutoML GBM	RMSE=703.81; 132M vehicles by 2028 at 1.9% CAGR
Q6	Python forecast replication	Holt-Winters smoothing	MAPE=2.89% — exceeds Transport for London 7% benchmark

Complete Model Performance Summary

Model	Platform	Algorithm	Key Metric	Performance	Production Ready
Regression (Bus Terminal)	H2O AutoML	GBM (best)	RMSE	703.81	YES — monthly retraining
Regression (Bus Terminal)	Python sklearn	Random Forest	RMSE / R²	752 / 0.9438	YES — audit fallback
Regression (Bridge & Tunnel)	Python sklearn	Random Forest	RMSE / R²	150.51 / 0.6657	YES
Classification	H2O AutoML	GBM_5 (best)	Mean per-class error	0.00074	YES — top model
Classification	Python sklearn	RF + SMOTE	Accuracy / F1	90.5% / 0.74 macro	YES — SMOTE deployed
Clustering (Bus Terminal)	H2O K-Means	K-Means k=3	B/T ratio	57.2%	YES — 3 operational segments
Clustering (Bridge & Tunnel)	Python sklearn	KMeans + Scaler	3 clusters	78.6% off-peak	YES — maintenance scheduling
Time Series	Python statsmodels	Holt-Winters	MAPE	2.89%	YES — 2028 forecast ready

Rank	Factor	SHAP Range	Business Interpretation
#1	Type (Bus vs Passenger)	Up to +8,000	Passenger events generate 3.6x more volume — fundamentally drives staffing
#2	Carrier Identity	±4,000	NJ Transit dominates with 687,758 total passengers
#3	Year	±1,000	Captures 46x post-COVID recovery (30,185 in 2020 → 1.39M in 2024)
#4	Month	±500	January peaks (386,223 cumulative); November troughs (260,530)
#5	Quarter	±300	Q1 consistently highest — enables proactive staffing 60-90 days ahead
Last	Speed variables	≈ 0	Avg_Speed, Delta_Speed_Efficiency all near zero — road conditions do NOT drive ridership

Business implication: Capital investment in road speed improvements will NOT drive ridership growth. Port Authority should prioritize carrier contract negotiations and service type expansion.

Q2: Toll Violations Analysis

Year	Total Violations	YoY Change	Business Context
2013	2,979,380	Baseline	Pre-cashless baseline — lowest violation year
2018	8,079,838	+35.3%	Rapid growth — EZPass expanding but violations rising
2020	12,740,778 (PEAK)	+50.2%	COVID-19: reduced enforcement, unmanned toll plazas
2021	11,613,671	-8.8%	Partial return to enforcement
2023	4,221,114	-55.7%	Full cashless tolling rollout
2024	4,296,383	Stable	New post-cashless structural baseline

Top violation facilities: GWB Lower (~19M) + GWB Upper (~18.5M) = **46% of all violations**. **Recommended action:** ALPR camera upgrade at these 2 facilities → **\$15-25M/year recovered**.

Time Period	Pattern	Operational Implication
06:00–09:00 weekdays	AM PEAK	Maximum toll plaza staffing required
15:00–18:00 weekdays	PM PEAK	Full cashier and incident response deployment
00:00–05:00 all days	Off-peak trough	Optimal maintenance window
Friday all day	+8-12% above Mon–Thu	Treat as separate peak-planning event
Saturday 10:00–18:00	Compressed midday peak	No AM spike — recreational travel
January	386,223 cumulative (highest)	Pre-position resources in late December
November	260,530 cumulative (lowest)	Maintenance scheduling window

Q4: Congestion Pricing 2025 — YoY Results

Facility	2024 Volume	2025 Volume	YoY Change
GWB Upper	27,552,657	11,509,746	-58.2%
Goethals	18,192,674	7,374,059	-59.5%
Outerbridge	14,945,108	5,918,453	-60.4%
GWB Lower	22,689,107	8,823,656	-61.1%
Holland Tunnel	15,812,944	6,061,907	-61.7%
Lincoln Tunnel	—	—	-62.0%
Bayonne	—	—	-62.4%

Key finding: Broad-based trip suppression — NOT route diversion. Mirrors London Congestion Zone 2003 (-15%) and Stockholm 2006 (-20%) outcomes.

Q5 & Q6: 2028 Traffic Forecast

Year	H2O GBM Forecast	Python Holt-Winters	Difference
2026	~127,500,000	126,990,408	<1% — strong model agreement
2027	~130,000,000	129,352,796	<1%
2028	~132,500,000	131,715,185	<1% — 1.9% CAGR
MAPE	N/A (RMSE=703)	2.89%	Exceeds TfL 7% benchmark

Traffic Clustering — Operational Segments

Cluster	Size	Avg Traffic	Avg Large Trucks	Business Profile
0 — Off-Peak Standard	4,230,436 (78.6%)	164.0	4.36	Optimal maintenance window
1 — Peak Auto Rush	901,561 (16.7%)	694.2	5.71	Maximum toll processing required
2 — Commercial Freight	251,381 (4.7%)	536.4	90.1	Highest pavement wear — priority inspection after Cluster 2 windows

Key insight: Commercial freight cluster (4.7% of records) causes disproportionate infrastructure damage. Large trucks cause 10,000–100,000x more pavement wear than autos.

Business Recommendations

Immediate Actions (0–6 Months)

Recommendation	Evidence	Estimated Impact
ALPR upgrade at GWB Lower & Upper	46% of all violations at these 2 facilities	\$15–25M/year recovered
EZPass conversion campaign	85.7% vs. 94% ISTHA benchmark	Eliminate ~2.7M cash transactions; \$8–15M/year
Expand maintenance windows 30–40%	All facilities down 58–62% in 2025; Cluster 0 = 78.6% of lane-hours	Reduce emergency repair costs 15–25%
Deploy SMOTE classification model	Low-class recall improved from 0% to functional	Automated service disruption warnings

Total Estimated Annual Savings: \$23–40M

Tech Stack

Languages:	Python 3.9, R
ML Platforms:	H2O AutoML (Flow), Python scikit-learn
ML Libraries:	statsmodels, imblearn (SMOTE), AutoTS, H2O GBM/DRF/DeepLearning
Feature Analysis:	SHAP (SHapley Additive exPlanations)
Visualization:	Power BI, matplotlib, seaborn
Cloud:	Google Cloud Platform, Vertex AI
Environment:	Jupyter Notebook, Anaconda
Data Scale:	5.38M bridge/tunnel records + 1,601 bus terminal records

```
PANYNJ-Traffic-Analytics/
|
├─ data/
|   ├── All_Recorded_Traffic.txt           # 5.38M bridge & tunnel records
|   └── Final_Dataset.csv                 # Bus terminal integrated
dataset (1,601 rows)
|   ├── train_data.csv / valid_data.csv   # 80/20 temporal train-test
split
|   └── Cleaned_Traffic_Stratified_100k.csv # Stratified sample for
modeling
|
├─ notebooks/
|   ├── Analysis_on_Traffic.ipynb         # Full EDA, Random Forest,
KMeans (Python)
|   └── Time_Series_Model.ipynb           # AutoTS + Holt-Winters
MAPE=2.89%
|   ├── H2O_Random_Forest.ipynb           # H2O AutoML regression
leaderboard
|   └── H2O_Classification_Model.ipynb     # H2O AutoML classification
(GBM best)
|   └── H2O_K_Means.ipynb                 # H2O K-Means clustering (k=3)
|
├─ dashboards/
|   ├── PANYNJ_Dashboard.html             # Interactive Power BI export
|   └── screenshots/                     # Dashboard page screenshots
|
├─ reports/
|   ├── PPR3_Final_Report.docx            # Full analytics report (April
2026)
|   ├── PPR2_Enhanced_Group3.docx         # PPR2 progress report
|   └── PPR3_Presentation.pptx           # Final presentation slides
|
└─ README.md
```

Key Code Snippets

Holt-Winters Forecast (Q6 — MAPE 2.89%)

```
python

from statsmodels.tsa.holtwinters import ExponentialSmoothing
from sklearn.metrics import mean_absolute_percentage_error

hw = ExponentialSmoothing(
    train_ts, trend='add', seasonal='add',
    seasonal_periods=12, initialization_method='estimated'
).fit(optimized=True)

val_pred = hw.forecast(len(test_ts))
mape = mean_absolute_percentage_error(test_ts, val_pred)
print(f"Holdout MAPE: {mape*100:.2f}%")
# Output: 2.89% — exceeds Tfl 7% industry benchmark
# Forecasts: 2026=126,990,408 | 2027=129,352,796 | 2028=131,715,185
```

```
python

from imblearn.over_sampling import SMOTE
from sklearn.ensemble import RandomForestClassifier

sm = SMOTE(random_state=42, k_neighbors=2)
X_sm, y_sm = sm.fit_resample(X_train, y_train)

rf = RandomForestClassifier(n_estimators=100, max_depth=15,
                           min_samples_leaf=5, random_state=42)
rf.fit(X_sm, y_sm)

# Before SMOTE: 95.6% accuracy, Low class recall = 0%
# After SMOTE: 90.5% accuracy, Low class: detectable (critical operational imp
```

Congestion Pricing Analysis (Q4)

```
python

vol_24 = df[df['Yr']==2024].groupby('FAC_B')['TOTAL'].sum()
vol_25 = df[df['Yr']==2025].groupby('FAC_B')['TOTAL'].sum()
yoy = pd.DataFrame({'2024': vol_24, '2025': vol_25}).dropna()
yoy['pct_change'] = ((yoy['2025'] - yoy['2024']) / yoy['2024'] * 100).round(2)
print(yoy.sort_values('pct_change'))

# Results: All facilities -58% to -62% – trip suppression confirmed (not divers
```

SHAP Feature Importance (Q1)

```
python

import shap
from sklearn.ensemble import RandomForestRegressor

rf_model = RandomForestRegressor(n_estimators=100, max_depth=15, random_state=4
rf_model.fit(X_train, y_train)

explainer = shap.TreeExplainer(rf_model)
shap_values = explainer.shap_values(X_test)
shap.summary_plot(shap_values, X_test)

# Top factors: Type > Carrier > Year > Month > Quarter
# Speed variables (Avg_Speed, Delta) ≈ 0 SHAP – no impact on ridership
```

How to Run

```
bash
```

```
# Clone the repository
git clone https://github.com/alekhya-kodhanda/PANYNJ-Traffic-Analytics.git
cd PANYNJ-Traffic-Analytics

# Install all dependencies
pip install pandas numpy matplotlib seaborn statsmodels scikit-learn \
    h2o autots imbalanced-learn shap jupyter

# Launch Jupyter
jupyter notebook

# Run notebooks in this order:
# 1. Analysis_on_Traffic.ipynb (EDA + Random Forest + KMeans)
# 2. H2O_Random_Forest.ipynb (H2O AutoML regression)
# 3. H2O_Classification_Model.ipynb (H2O AutoML classification)
# 4. H2O_K_Means.ipynb (H2O clustering)
# 5. Time_Series_Model.ipynb (Holt-Winters forecast)
```

Global Benchmarking

Organization	Technique	Benchmark	Our Result
Transport for London	Time-series forecasting	7% MAPE	2.89% MAPE
Chicago ISTHA	EZPass adoption	94% adoption	85.7% (gap = opportunity)
Singapore LTA	ML congestion pricing	30% congestion reduction	58-62% reduction observed
Hong Kong MTR	RF + SARIMA hybrid	MAPE <3.2%	2.89%
New York MTA	Prophet + GBM	MAPE <6%	2.89%

Team

Built by **Group 3** — MS Business Analytics, University of New Haven (Spring 2026)

Member	Role	Contribution
Alekhya Kodhanda	Classification Lead	SMOTE fix, H2O AutoML classification, confusion matrix analysis
Imran Hossain Rasel	Regression & SHAP Lead	SHAP analysis, Q1 top factors, regression model improvement
Nishanth Seenivasakap	Clustering Lead	K-Means clustering, cluster visualization, segmentation
Jasmine Bajracharya	Q4 Congestion Analysis	Congestion pricing 2025, facility diversion analysis, EDA
Simranpreet Kaur	Forecast & Integration Lead	Time series forecast, report assembly, business recommendations

Connect

Alekhya Kodhanda — Classification Lead | MS Business Analytics @ University of New Haven [LinkedIn](#) · [GitHub](#) Open to full-time Data Analyst / BI Analyst / Data Science roles in Connecticut and beyond.